Candidate Seat no………….

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech (EC) Examination

Nov-Dec. 2018

EC343 Digital Signal Processing

Date: 24.11.2018, Saturday **Time: 10.00 a.m. To 01.00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 For each of the following systems, determine whether the system is static/dynamic, causal/non-causal, linear/non-linear, time invariant/time variant. **[06]**

(i) $y(n) = x(n) + 3u(n+1)$

(ii) $y(n) = x(an)$ where 'a' is any integer greater than 1

Q – 2.a Determine the response of the system whose input and impulse response are given as follows, Use Graphical method. **[05]**

$$x(n) = u(n+1) - u(n-4) - \delta(n-5)$$
$$h(n) = [u(n+2) - u(n-3)] \cdot (3 - |n|)$$

Q – 2.b **Answer any two questions.** **[10]**

(i) State sampling theorem. For an analog signal,

$$x(t) = 3\cos 600\pi t + 2\cos 1800\pi t$$

Calculate Nyquist rate.

(ii) Determine the homogeneous solution of the system described by:

$$y(n) - 3y(n-1) - 4y(n-2) = x(n)$$

(iii) Define correlation and determine the cross correlation for the following sequences:

$$x_1(n) = \{1, 2, 3, 4\}$$
$$x_2(n) = \{4, 3, 2, 1\}$$

Q - 3 **Answer any two questions.** **[14]**

a. Determine the Z- transform of following sequence:

$$x(n) = \left(\frac{-1}{3}\right)^n u(n) - \left(\frac{1}{2}\right)^n u(-n-1)$$

b. Define Region of Convergence (ROC). Enlist properties of Region of Convergence.

c. Find Inverse Z transform using long division method

$$X(Z)=\frac{1}{(1+z^{-1})(1-z^{-1})^2}$$

ROC: $|Z|>1$

SECTION – II

Q - 4.a Determine Linear Convolution using DFT for the following Sequence. **[03]**

$x(n)=\{1,2\}$ and h(n)={1,2,3}

b. The five samples of the 9-point DFT are given as follows: **[04]**

$$X(0)=23$$
$$X(1)=2.242-j$$
$$X(4)=-6.379+j4.121$$
$$X(6)=6.5+j2.59$$
$$X(7)=-4.153+j0.264$$

Determine the remaining samples of DFT if the corresponding time domain sequence is real.

Q – 5.a Compare Infinite Impulse Response and Finite Impulse Response filter. **[04]**

Q – 5.b Answer any two questions. **[10]**

(i) Draw the structures of the following discrete time system:

$$H(z)=\frac{(1+z^{-1})^2}{1-0.75z^{-1}+0.125z^{-2}}$$

(i) Direct form- I **(ii)** Direct form- II

(iii) Cascade form **(iv)** Parallel form

(ii) Explain Impulse Invariant Method to design Infinite Impulse Response Filter.

(iii) Evaluate Circular Convolution of

$$x(n)=u(n)-u(n-4)$$
$$h(n)=u(n)-u(n-3)$$

Assuming N=4. Show calculations by appropriate usage of equations and relevant sketches.

Q – 6.a Prove following properties of Twiddle factor: **[07]**

(i) $W_N^k=W_N^{k+N}$

(ii) $W_N^{k+\frac{N}{2}} = -W_N^{k+N}$

(iii) $W_N^2 = W_{N/2}$

b. Obtain 8 point DFT for the given sequence using DIF FFT algorithm. **[07]**

$$x(n) = n+1$$

OR

b. Given $x(n) = \{0,1,2,3\}$ find 4 point DFT using FFT algorithm and also obtain inverse FFT to verify the result. **[07]**